The empirical recurrence for OEIS A229377, recovered from its tail

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Abstract

OEIS A229377 records "Empirical recurrence of order 64" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 74$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 66$. Its last coefficient is $c_{64} = -128053272000 \neq 0$, so no further tail satisfies anything shorter; any order-64 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A229377 is "Number of nX5 0..2 arrays avoiding 11 horizontally, 22 vertically and 00 diagonally or antidiagonally.". It has offset 1 and begins

$$164, 2138, 48490, 956866, 20014278, 407900408, 8454015792, 173331549156, \ldots$$

The entry states:

> Empirical recurrence of order 64 (see link above)

As of the "Last modified" line on the live entry (Oct 04 2025 00:10:23, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 74$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 74$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 64: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 64.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 148$ exact terms from $n = 2$ on, it returns order 64 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{64} c_i\, a(n-i),$$

c_1	14	c_{17}	14716067314316312	c_{33}	11264708506522434343610	c_{49}	97099403324
c_2	689	c_{18}	22760128419027857	c_{34}	842847290456170661195	c_{50}	−986820259
c_3	−8484	c_{19}	−198880253762991873	c_{35}	−20192237202159481366065	c_{51}	−2025953646
c_4	−181798	c_{20}	−225867101273777780	c_{36}	−764347842345018782731	c_{52}	528467230
c_5	2006948	c_{21}	2037863093462778624	c_{37}	28394143340004629588204	c_{53}	3020486439
c_6	24876511	c_{22}	1688651806169396623	c_{38}	349695503405708630435	c_{54}	189282707
c_7	−251598942	c_{23}	−16009528316740316963	c_{39}	−31215486693926528419680	c_{55}	−311647514
c_8	−2024631106	c_{24}	−9573626362870247681	c_{40}	137407453087569686013	c_{56}	−50073166
c_9	19148382820	c_{25}	97261767271634509570	c_{41}	26691010322115224757964	c_{57}	212773350
c_{10}	106172449737	c_{26}	41300158000619709800	c_{42}	−380524103727384454281	c_{58}	56642663
c_{11}	−956664667403	c_{27}	−459962425807939119885	c_{43}	−17626201158221908738815	c_{59}	−8977902
c_{12}	−3778887188431	c_{28}	−135560237746698303487	c_{44}	330049390155312806092	c_{60}	−3414183
c_{13}	32998462468738	c_{29}	1701416828362434061575	c_{45}	8908283827284120133567	c_{61}	20752487
c_{14}	94629854442022	c_{30}	336880517529947148748	c_{46}	−168774239409679475738	c_{62}	1049821
c_{15}	−813771184823021	c_{31}	−4938383416555603072026	c_{47}	−3406088262337544398176	c_{63}	−197296
c_{16}	−1710922748921483	c_{32}	−625620464251027962905	c_{48}	53931616553082505177	c_{64}	−12805

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 66$, and it is the recurrence OEIS A229377 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{64} - c_1 x^{64-1} - \cdots - c_{64}$. Its constant term is $-c_{64} = 128053272000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 64 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 64, so $\deg q = 64$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 64, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 64, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -128053272000.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-64 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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